Points to Remember

- Random variable is a real valued function defined on the sample space. It is always associated with the outcomes of the random experiment.
- Random variables are classified based on the nature of the range of values taken by it.
- Probability function is used to find the probabilities at the selected point or at the selected interval, in the range of the random variable.
- Cumulative distribution functions are used to find the probabilities in the subset of the range of the random variable.
- Summary measures of probability distribution (Mean and Variance) are obtained through the three approaches: (i) expectation (ii) m.g.f and (iii) characteristic function.

EXERCISE 9

BCYT5M

I . Choose the best answer:

1. $f(x) = \begin{cases} kx^2 & 0 \le x \le 3 \\ 0 & elsewhere \end{cases}$ is a pdf then the value of K is

(a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{9}$ (d) $\frac{1}{12}$

2. A random variable X has the following probability distribution

X	0	1	2	3	4	5
$P(X = x)$	$\frac{1}{4}$	$2a$	$3a$	$4a$	$5a$	$\frac{1}{4}$

Then $P(1 \le X \le 4)$ is

(a) $\frac{10}{21}$ (b) $\frac{2}{7}$ (c) $\frac{1}{2}$ (d) $\frac{1}{14}$

3. X is a discrete random variable which takes the value 0, 1, 2 and $P(X = 0) = \frac{144}{169}$; $P(X=1) = \frac{1}{169}$ and $P(X=2) = K$, then the value of K is

(a) $\frac{145}{169}$ (b) $\frac{24}{169}$ (c) $\frac{2}{169}$ (d) $\frac{143}{169}$

4. Given $E(X+C) = 8$ and $E(X-C) = 12$ then C is equal to

(a) -2 (b) 4 (c)-4 (d) 2